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McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 23
Reduction of order
Thursday 5 November 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Reduction of order

[Trench §5.6]

Finding a second independent solution

Consider the most general homogeneous second-order linear ODE

$$L[y] \;=\; a_2(x)y'' + a_1(x)y' + a_0(x)y \;=\; 0\,, \qquad (\heartsuit)$$

where the coefficient functions are continuous on an interval $\mathcal{I}$, with $a_2(x) \neq 0$ on $\mathcal{I}$ (the functions $a_i(x)$ need not be constant).

Suppose we have found *only one* solution of Eq. ($\heartsuit$), say $y_1 \not\equiv 0$.

- We need a second solution y_2, linearly independent of y_1.
- If we can find such a y_2, then $\{y_1, y_2\}$ is a fundamental set of solutions, and the general solution is

$$y(x) \;=\; c_1 y_1(x) + c_2 y_2(x).$$

- *Can we find such a solution y_2 using the solution y_1 we already know?*

Variation of parameters with one known solution

Every constant multiple of the known solution is also a solution:

$$y(x) = C\,y_1(x), \qquad C \in \mathbb{R}.$$

These are all multiples of y_1; none gives us the second linearly independent solution we need.

Use variation of parameters: replace the constant C by a function $u(x)$, and try

$$\boxed{y(x) = u(x)y_1(x)}$$

We must find $u(x)$ so that y satisfies the ODE. To obtain a second independent solution, u must be nonconstant.

When deriving $u(x)$, we will use colour to indicate functions that are given and suppress dependence on x (*e.g.*, a_2 and y_1 to mean $a_2(x)$ and $y_1(x)$).

Note: Trench [Eqs. (5.6.1)–(5.6.3)] writes P_0, P_1, P_2 for our a_2, a_1, a_0, respectively.

Why does this reduce the order?

With $y = uy_1$, the product rule gives

$$\begin{aligned} y' &= u'y_1 + uy_1', \\ y'' &= u''y_1 + 2u'y_1' + uy_1''. \end{aligned}$$

Substitute into $a_2y'' + a_1y' + a_0y = 0$ and collect terms:

$$\begin{aligned} &a_2y_1u'' + (2a_2y_1' + a_1y_1)u' \\ &\quad + \underbrace{(a_2y_1'' + a_1y_1' + a_0y_1)}_{L[y_1]=0}u = 0. \end{aligned}$$

Thus the term involving u disappears:

$$\boxed{a_2y_1u'' + (2a_2y_1' + a_1y_1)u' = 0} \qquad \text{(R)}$$

This equation contains u' *and* u'', *but not* u, so it is a ***first order linear ODE for*** u'.

[Trench Eqs. (5.6.4)–(5.6.5)]

Reduction of order

In (R), set $\boxed{z = u'}$ so $z' = u''$, giving

$$a_2 y_1 z' + (2a_2 y_1' + a_1 y_1) z = 0.$$

Divide by $a_2 y_1$ to obtain

$$\boxed{z' + \left(2\frac{y_1'}{y_1} + \frac{a_1}{a_2}\right) z = 0,} \qquad (\spadesuit)$$

a *homogeneous first-order linear ODE* for z.
Note: ***We must restrict to a subinterval of $\mathcal{I}$ on which $y_1 \neq 0$.***

1. Solve for z using EMSID or separation of variables. *Note:* We must choose a *nonzero* solution z (the solution $z \equiv 0$ would not lead us to a second linearly independent solution of Eq. ($\heartsuit$)).
2. Integrate $u' = z$ to find a nonconstant u.
3. Then $y_2 = uy_1$ is a second solution. It is independent of y_1 because $y_2/y_1 = u$ is nonconstant.

How does reduction of order affect initial conditions?

Suppose the initial conditions are $y(x_0) = y_0$ and $y'(x_0) = v_0$.

Since $y = uy_1$, we have

$$y_0 = u(x_0)y_1(x_0) \quad \Longrightarrow \quad \boxed{u(x_0) = \frac{y_0}{y_1(x_0)}}$$

Since $y' = u'y_1 + uy_1'$ and $z = u'$, we also have

$$v_0 = z(x_0)y_1(x_0) + u(x_0)y_1'(x_0),$$

so

$$\boxed{z(x_0) = \frac{v_0 - u(x_0)y_1'(x_0)}{y_1(x_0)}}$$

This is what we need to make Eq. (♠) an IVP.

Alternatively: find the general solution for y first, then use the original initial conditions to determine its constants.

Example: a homogeneous IVP

Example $((3x-1)y'' - (3x+2)y' - (6x-8)y = 0,$ $y(0) = 2, \quad y'(0) = 3$ [Trench §5.6, Exercise 32])

Use the known solution $y_1 = e^{2x}$.

Choose interval $\mathcal{I}$ on which $3x - 1 \neq 0$: take $\mathcal{I} = (-\infty, 1/3)$.

Use variation of parameters: $y = ue^{2x}$. Then

$$y' = (u' + 2u)e^{2x}, \qquad y'' = (u'' + 4u' + 4u)e^{2x}.$$

Substitution and cancellation of the u terms give

$$e^{2x}\left[(3x-1)u'' + (9x-6)u'\right] = 0.$$

Set $z = u'$ and divide by $(3x-1)e^{2x}$:

$$z' + \frac{9x-6}{3x-1}z = 0 \quad \Longleftrightarrow \quad z' + \left(3 - \frac{3}{3x-1}\right)z = 0.$$

Continued...

Example: a homogeneous IVP

Example $((3x-1)y'' - (3x+2)y' - (6x-8)y = 0,$ $y(0)=2, \quad y'(0)=3$ [Trench §5.6, Exercise 32])

Separate variables in $z' + \left(3 - \frac{3}{3x-1}\right) z = 0$, choosing $z \neq 0$:

$$\begin{aligned} \frac{z'}{z} &= -3 + \frac{3}{3x-1}, \\ \ln|z| &= -3x + \ln|3x-1| + \text{constant}, \\ |z| &= A|3x-1|e^{-3x}, \qquad A = e^{\text{constant}} > 0. \end{aligned}$$

On $x < 1/3$, we have $|3x-1| = 1-3x$. Since z is continuous and never zero on $x < 1/3$, its sign is fixed. Absorb its sign into the constant: $C = \pm A$.

$$\boxed{z = C(1-3x)e^{-3x}}$$

Continued...

Example: a homogeneous IVP

Example $((3x-1)y'' - (3x+2)y' - (6x-8)y = 0,$ $y(0) = 2, \quad y'(0) = 3$ [Trench §5.6, Exercise 32])

Now $u' = z = C(1-3x)e^{-3x}$. Integrate by parts:

$$\begin{aligned}\int (1-3x)e^{-3x}\,dx &= -\tfrac{1}{3}(1-3x)e^{-3x} - \int e^{-3x}\,dx \\ &= -\tfrac{1}{3}(1-3x)e^{-3x} + \tfrac{1}{3}e^{-3x} + \text{constant} \\ &= xe^{-3x} + \text{constant}.\end{aligned}$$

Alternatively, notice directly that $(xe^{-3x})' = (1-3x)e^{-3x}$. Thus

$$u = Cxe^{-3x} + D \quad \Longrightarrow \quad y = De^{2x} + Cxe^{-x}.$$

The initial conditions give $D = 2$ and $2D + C = 3$, so

$$\boxed{y = 2e^{2x} - xe^{-x}, \qquad x < 1/3}$$

Where did the second solution come from?

In the homogeneous example we just did, reduction of order gave

$$y = De^{2x} + Cxe^{-x}.$$

- A constant choice of u in $y_2 = uy_1$ would have given a multiple of the solution we already knew: $y_1 = e^{2x}$.
- The nonconstant choice $u = xe^{-3x}$ gives the second solution $y_2 = xe^{-x}$.
- We could compute their Wronskian to check independence. Here it is easier to note that y_2 is not a constant multiple of y_1 on $(-\infty, 1/3)$.

Thus we found a second independent solution *without having to guess its form*.

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Reduction of order: constants and independence**
- Submit.

Poll recap: constants and independence

We know a solution $y_1(x) \neq 0$ of a homogeneous second-order linear ODE. To reduce the order, we use variation of parameters: $y = uy_1$, followed by $z = u'$. Which statements are correct?

1. Choosing $z \equiv 0$ gives only the zero solution $y \equiv 0$.
   False. $u' = 0$ gives $u = C$, so $y = Cy_1$, which need not be zero.

2. Choosing $z \equiv 0$ gives all constant multiples of y_1.
   True. The constant C is arbitrary, giving every multiple of y_1.

3. A nonzero solution of the reduced equation for z gives a solution y linearly independent of y_1.
   True. $u' = z \not\equiv 0$ makes u nonconstant, so $y = uy_1$ cannot be a constant multiple of y_1.

4. When finding the general solution, we can omit the integration constant when recovering u from $u' = z$.
   False. The constant D in $u = \int z\,\mathrm{d}x + D$ contributes Dy_1 to y. We need both arbitrary constants.

Poll

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- Fill in poll **Reduction of order with constant coefficients**
- Submit.

Variation of parameters with forcing

Now consider $L[y] = g$, where g is continuous on $\mathcal{I}$. The homogeneous equation $L[y] = 0$ is its ***complementary equation***. Suppose we have found one solution y_1 of the complementary equation, but not yet a second independent solution.

Use the same variation of parameters substitution $y = uy_1$. The cancellation that gave (R) now gives

$$a_2 y_1 u'' + (2a_2 y_1' + a_1 y_1)u' = g.$$

- Setting $z = u'$ again gives a first-order linear equation, now with a nonzero right-hand side.
- Solving for z and then integrating gives two arbitrary constants, as expected for a second-order ODE.
- We obtain the complementary and particular parts together, rather than finding them separately.

Note: Trench [Eq. (5.6.1)] writes F for our g.

Example: a nonhomogeneous IVP

Example ($4x^2y'' - 4x(x+1)y' + (2x+3)y = 4x^{5/2}e^{2x}$, $y(1) = 0, \quad y'(1) = 0$ [Trench §5.6, Exercise 16; initial conditions added])

Use the known complementary solution $y_1 = \sqrt{x}$. $x > 0$.

Use variation of parameters: $y = \sqrt{x}\,u$. Then

$$y' = \sqrt{x}\,u' + \frac{u}{2\sqrt{x}}, \qquad y'' = \sqrt{x}\,u'' + \frac{u'}{\sqrt{x}} - \frac{u}{4x^{3/2}}.$$

Substituting and collecting terms gives

$$4x^{5/2}(u'' - u') + \sqrt{x}\,\underbrace{[-1 - 2(x+1) + (2x+3)]}_{0}\,u = 4x^{5/2}e^{2x}.$$

Divide by $4x^{5/2}$ and set $z = u'$:

$$u'' - u' = e^{2x} \quad \Longrightarrow \quad \boxed{z' - z = e^{2x}}$$

Continued...

Example: a nonhomogeneous IVP

Example ($4x^2y'' - 4x(x+1)y' + (2x+3)y = 4x^{5/2}e^{2x}$, $y(1) = 0, \quad y'(1) = 0$ [Trench §5.6, Exercise 16; initial conditions added])

Solve $z' - z = e^{2x}$ by EMSID:

$$\begin{aligned} \mu &= e^{-x}, \\ (e^{-x}z)' &= e^{x}, \\ e^{-x}z &= e^{x} + C, \\ z &= e^{2x} + Ce^{x}. \end{aligned}$$

Integrate $u' = z$:

$$u = \tfrac{1}{2}e^{2x} + Ce^{x} + D.$$

By the initial conditions,

$$u(1) = 0, \qquad z(1) = 0.$$

Hence

$$\begin{aligned} e^2 + Ce &= 0, \\ \tfrac{1}{2}e^2 + Ce + D &= 0, \end{aligned}$$

giving $C = -e, \quad D = e^2/2$.
Finally, $y = \sqrt{x}\,u$:

$$\boxed{y = \frac{\sqrt{x}}{2}(e^x - e)^2}$$

Example: a nonhomogeneous IVP

Example ($4x^2y'' - 4x(x+1)y' + (2x+3)y = 4x^{5/2}e^{2x}$, $y(1) = 0, \quad y'(1) = 0$ [Trench §5.6, Exercise 16; initial conditions added])

Alternatively: use the original conditions and the product rule:

$$y = \sqrt{x}\left(\tfrac{1}{2}e^{2x} + Ce^x + D\right),$$

$$y' = \frac{\frac{1}{2}e^{2x} + Ce^x + D}{2\sqrt{x}} + \sqrt{x}\left(e^{2x} + Ce^x\right).$$

From $y(1) = 0$:

$$\tfrac{1}{2}e^2 + Ce + D = 0.$$

Then $y'(1) = 0$ gives

$$0 = \tfrac{1}{2}\underbrace{\left(\tfrac{1}{2}e^2 + Ce + D\right)}_{0 \text{ since } y(1)=0} + e^2 + Ce.$$

Thus $C = -e$, and then $D = e^2/2$. Hence

$$\boxed{y = \frac{\sqrt{x}}{2}(e^x - e)^2, \qquad x > 0}$$

Suggested practice problems

Trench §5.6: Reduction of Order

- *Homogeneous equations: find a second independent solution*
  Exercises 18, 19, 22, 25.
- *Nonhomogeneous equations: find the general solution*
  Exercises 1–3, 5, 8, 10.
- *Initial value problems*
  Exercises 31, 33.

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)\,y = f(x) \qquad (*)$$

EMSID *method to find the general solution:*

1. **E**valuate the integrating factor $e^{\int P(x)\,dx}$
2. **M**ultiply both sides of $(*)$ by the integrating factor
3. **S**implify using product rule:

$$\frac{d}{dx}\left(e^{\int P(x)\,dx} y\right) = e^{\int P(x)\,dx} f(x) \qquad (**)$$

4. **I**ntegrate both sides of $(**)$ (but add only one constant).
5. **D**ivide both sides by the integrating factor (to isolate y).

Note: "EMSID" isn't standard terminology; it's just a suggestion to help you remember the method.